

# BTC/USD Trading Strategy

## Backtest Report

### Strategy Logic & Hypothesis

**So, my working hypothesis was that BTC/USD doesn't just move randomly—it has these massive trending phases interrupted by super choppy, sideways action. A basic breakout script would get absolutely chopped to pieces in the sideways markets. To fix this, I stacked a couple of filters on a 20-day Donchian channel breakout.**

**First, the breakout has to align with the overall trend direction using a 20/50 EMA crossover (fast above slow for going long, vice versa for shorts). Second, I threw in an ADX (14) > 20 condition to ensure the trend actually has some momentum and isn't just drifting. For exits, fixed targets are kind of mid, so I coded a 3x ATR trailing stop. This lets the winners run while cutting the losers early. The script can also hard-flip from long to short if a solid opposite breakout triggers before the stop hits.**

### Performance Metrics

**I ran the backtest on daily BTC/USD data from 2018 to 2022. I started with a \$1,000 hypothetical portfolio, compounded it, and factored in a 0.15% broker fee per trade to keep it realistic. I also sanity-checked the code by running it signal-by-signal on truncated data, so there is zero lookahead bias**

| Metric                         | Value                    |
|--------------------------------|--------------------------|
| Total Trades                   | 23                       |
| Winning / Losing Trades        | 13 / 10                  |
| Win Rate                       | 56.5%                    |
| Net Profit (on \$1000)         | \$8,909.90               |
| Benchmark Return (Buy & Hold)  | 23.6% (\$236.35)         |
| Sharpe Ratio                   | 1.15                     |
| Maximum Drawdown               | 37.65%                   |
| Average Drawdown               | 7.41%                    |
| Largest Win / Largest Loss     | \$3,402.03 / -\$1,448.56 |
| Winning Streak / Losing Streak | 4 / 2                    |
| Average Holding Time           | 46 days                  |

**The bot massively outperformed just holding Bitcoin during this training window. Compounding the account on a few massive trending moves really amplified the returns. That being said, a 37.65% max drawdown is pretty rough, and 23 trades over 5 years is a super small sample size, so I'm treating these numbers with caution before throwing live money at it.**

## Parameter Selection

The two main dials I had to tweak were the ADX threshold (how strong the trend needs to be) and the ATR multiplier (how much breathing room the trailing stop gets). Instead of just guessing, I wrote a script to sweep both and log the results

- Pushing ADX to 25 hurt performances across the board—it filtered out way too much action and only caught the tail end of moves.
- An ADX of 20 with a 3x ATR stop gave the best Sharpe (1.15) and maxed out the profit.
- A 2x ATR stop was too tight (got stopped out too early), and a 4x gave a brutal drawdown without boosting the Sharpe. So, 3x ATR + 20 ADX was the final pick.

| ATR Mult | ADX Th. | Trades | Sharpe | Net Profit | Max DD | Note                  |
|----------|---------|--------|--------|------------|--------|-----------------------|
| 2.0      | 15      | 36     | 1.05   | \$6,012    | 29.3%  |                       |
| 2.0      | 20      | 33     | 1.01   | \$5,252    | 29.3%  |                       |
| 2.0      | 25      | 26     | 0.52   | \$888      | 51.0%  | ADX filter too strict |
| 3.0      | 15      | 26     | 1.08   | \$7,567    | 37.7%  |                       |
| 3.0      | 20      | 23     | 1.15   | \$8,910    | 37.7%  | Best                  |
| 3.0      | 25      | 21     | 0.42   | \$546      | 59.0%  | ADX filter too strict |
| 4.0      | 15      | 22     | 1.07   | \$8,377    | 49.2%  |                       |
| 4.0      | 20      | 20     | 1.02   | \$7,038    | 49.2%  |                       |
| 4.0      | 25      | 18     | 0.45   | \$606      | 74.3%  | ADX filter too strict |

## Limitation

Ngl, breaking down the net profit year-by-year shows the bot actually took a slight L in 2018 (-\$233.88) and did okay, but not amazing, in 2019-2020 (+\$731.50, +\$597.60). Honestly, almost all

**the crazy profit came from the massive 2021 bull run (+\$5,744.77), with 2022 padding it a bit more (+\$2,069.92). So, that sick 1.15 Sharpe ratio is heavily carried by one insane trending year. If we forward-test this on a period without a 2021-style mega-trend, the numbers might look way weaker.**

**Also, 23 total trades is a tiny sample size, so one lucky trade skews the overall stats heavily. Plus, since it's inherently a trend-following bot, it's still going to bleed in sideways markets. The ADX filter helps reduce trash entries, but it definitely doesn't eliminate them completely**